

The Propagation Framework

How Reality Derives Itself from Three Axioms

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The Claim

Everything is propagation. Matter is stable self-reinforcing wave patterns. Forces are refraction. Time is what propagation feels like from inside. Three axioms about how disturbances move through a medium derive the generation structure of the Standard Model, the Koide mass ratio, the Weinberg angle, and gravity - with zero free parameters.

Three Axioms

Axiom 1: Propagation is Fundamental

Everything that exists propagates. The medium is not empty space but a field capable of carrying a signal.

Axiom 2: Finite Causal Velocity

Every medium has a maximum signal speed c . No causal influence propagates faster.

Axiom 3: Coherence

Stable structure requires self-reinforcing, coherent propagation. Incoherent modes disperse. Among coherent states, the fundamental mode has minimal topological winding (Axiom 3b).

What the Axioms Derive

Result	Status	Key Fact
Three generations of matter	DERIVED (0.98)	$N=3$ is the unique integer satisfying $Q(N)=2/3$ in 3D topology
Topological weights (2,1)	DERIVED (0.98)	Fermion/boson distinction from $\pi_1(SO(3)) = \mathbb{Z}_2$
Koide ratio $Q = 2/3$	DERIVED (0.95)	Geometric identity: 120-degree spacing forces $R/A = \sqrt{2}$
Gravity as refraction	DERIVED (0.95)	Randers/Finsler equivalence. Shapiro delay: 0.01% error
8-hour sleep constant	DERIVED (0.92)	$Q=2/3$ of 24h cycle = 8h consolidation
Weinberg angle $\sin^2(\theta_W)$	DERIVED (0.90)	0.22310 via Axiom 3b. Matches PDG to 0.13 sigma
QCD confinement from λ_c	DERIVED (0.85)	Confinement radius via RG running. 1-loop: 2.2 fm vs 0.9 fm

The God Equation (Argued, 0.75)

$$\lambda_c = \sqrt{2} * l_P * \exp(4\pi^2 * N^{(D/2)} / b_0)$$

Predicts the top quark Compton wavelength from the Planck length alone. $N=3$, $D=3$, $b_0=16/3$. Predicted: 1.145×10^{-18} m. Measured: 1.14×10^{-18} m. Error: 0.4%. Zero free parameters. The bridge from internal phase closure to spatial coherence volume $N^{(D/2)}$ remains the key open theorem.

How to Kill It

Find a 4th generation particle at any mass, any coupling. The framework says it cannot exist - the exclusion is topological, not kinematic.

Show Koide $Q \neq 2/3$ for neutrinos. JUNO (2026) will constrain this. $>5\%$ deviation limits the framework's universality.

EEG phase transitions: If Critical Slowing Down does not precede cognitive insight in $\geq 7/10$ sessions, the cross-scale claim fails.

Tau g-2 at Belle II: Framework predicts a specific torsion correction. Pure QED agreement to 10^{-5} falsifies it.

What Failed (Honesty)

The harmonic series mass claim failed (CV=0.94, noise). The ϕ^3 electron/up ratio is interesting but uncertainty-limited ($p=0.007$). The spin-pair selection strike produced three independent no-go results. All documented in `sandbox_results.md`. A framework that only publishes successes is not science.

github.com/gwelby/propagation-framework | Clone and verify: `python propagation.py` | Full claims matrix: `CLAIMS.md` | Open gaps: `CONTRIBUTING.md`

This might be wrong. That is the point. The framework that survives contact with data is the one worth keeping.